

BIO202

CURRENT FINALTERMS 2023

MARYAM SALEEM KHAWAJA

MERRY  

VU EXPERT ZOOLOGISTS

WelcOme vularians....(◡_◡)

Subjective 2023 😞

1.characteristics of VLDLs

VLDLs (very low density lipo proteins) are assembled in the liver.

- composed predominantly of TAGs synthesized in liver and
- contain some cholesterol and cholesteryl esters
- As VLDL pass through the circulation, TAG is degraded and taken up by

peripheral tissues in the form of fatty acids, causing the VLDL to decrease in

size and become denser, called VLDL remnant.

2.primary structure of DNA

The primary structure of a nucleic acid is its covalent structure and nucleotide sequence.

- The back bone of the primary structure is the linear strand made by sugar phosphate residues, linked together, while the bases project laterally. This way a long, un-branched

chain is formed. Primary structure is a huge linear polymer of dNTPs that are joined to each other by 5'-

3' PDE bonds.

- The resulting long, un-branched chain has polarity.

- Both 5'-end and 3'-end are free. at 5'-end there is a free phosphate. at 3'-end there is a free

OH that are not attached to other nucleotides.

- Purines and pyrimidines project laterally from the backbone and forms a variable part. • The

variable part is concerned with the expression of genetic information. By convention, the structure of a single strand of nucleic acid is always written with the 5' end at the left and the 3' end at the right • that is, in the 5' to 3' direction

. • Some simpler representations of this penta deoxy ribonucleotide are • pA-C-G-T-A OH, •

pApCpGpTpA, and finally 5'-ACGTA-3'.

174 Nucleotides and Nucleic Acids-DNA

3. how activation energy effect rate of reaction

Activation energies, $G^\ddagger$, • The activation energies, $G^\ddagger$, for the • $S \rightarrow P$

and $P \rightarrow S$ reactions are indicated. • G'^0 is the overall standard free-

energy change in the direction $S \rightarrow P$

- The rate of a reaction reflects this activation energy: a higher activation energy corresponds to a slower reaction.
- Reaction rates can be increased by raising the temperature, thereby increasing the number of molecules with sufficient energy to overcome the energy barrier.
- Alternatively, the activation energy can be lowered by adding a catalyst.
- Catalysts enhance reaction rates by lowering activation energies
- The role of enzymes is to accelerate the interconversion of S and P.
- i.e enzymes lower the energy of activation, $\Delta G^\ddagger$, of a reaction.
- The enzyme is not used up in the process, and the equilibrium point is unaffected.

4. modified or Unusual bases

These unusual bases(also called modified bases) include

- Dihydrouracil • Pseudouridine
- Thymine.

5. functional group btana tha purine or pyrimidine ?

The most important functional groups of pyrimidines and purines are • ring

nitrogens, • carbonyl groups, and • exocyclic amino groups.

6. globosides

Globosides

- are glycosphingolipids
- with two or more sugars, usually
- D -glucose,
- D- galactose, or
- N-acetyl-D-galactosamine.

A glycosphingolipid that has only one sugar as the side chain is called a cerebroside

- Globosides are highly abundant in RBCs.
- Globosides are also found in human serum, spleen and liver.

Buffer

Buffers are aqueous systems that tend to resist changes in pH when small amounts of

acid (H⁺) or base (OH⁻) are added

- A buffer system consists of a weak acid (the proton donor) and its conjugate base (the proton acceptor) an example, a mixture of acetic acid and acetate ion, is a buffer system, If more H^+ is added to this solution, it simply shifts the equilibrium to the left, absorbing H^+ , so the $[H^+]$ remains unchanged.
- If H^+ is removed (e.g. by adding OH^-) then the equilibrium shifts to the right, releasing H^+ to keep the pH constant .
- Notice that the titration curve of acetic acid has a relatively flat zone

Buffer Composition:

To relatively the maintaining the pH of the solution , a buffer is must consist the acid-base pair meaning

either:

- i) A weak base and a conjugate acid.
- ii) A weak acid and conjugate base .

The use of one or more ,depend upon the desired pH when preparing the buffer.

Names of inorganic cofactors

If the non-protein moiety is a metal ion such as Zn^{2+} or Fe^{2+}

, it is called a cofactor.

Some Inorganic Elements That Serve as Cofactors for Enzymes

Cu^{2+}	Cytochrome oxidase
Fe^{2+} or Fe^{3+}	Cytochrome oxidase, catalase, peroxidase
K^{+}	Pyruvate kinase
Mg^{2+}	Hexokinase, glucose 6-phosphatase, pyruvate kinase
Mn^{2+}	Arginase, ribonucleotide reductase

Sterols

Steroids with eight to ten carbon atoms in the side chain at C-17 and a hydroxyl group at C-3 are classified as sterols. Sterols are major structural elements of biological membranes. Sterols are the major structural elements of biological membranes.

PI of histidine.

The pI of Histidine, with two groups that are positively charged when protonated, is 7.59 (the average of the pKa values of the amino and imidazole groups)

- Only histidine has an R group (pKa = 6.0) providing significant buffering power near the neutral pH usually found in the intracellular and extracellular fluids of most animals

Reaction velocity

The rate or velocity of a reaction (v) is the number of substrate molecules converted to product per unit time;

- Velocity is usually expressed as μmol of product formed per minute.
- The rate of an enzyme-catalyzed reaction increases with substrate concentration until a maximal velocity (V_{max}) is reached- reflecting the saturation with substrate of all available binding sites are occupied on the enzyme molecules present.

Behaviour of waxes in nature

Waxes are widely distributed in nature such as the secretion of certain insects as;

- Bees-wax, • Spermaceti of the sperm whale
- Waxes also form protective coatings of the skins and furs of animals and • leaves and fruits of plants.
- Triacontanol is a fatty alcohol of the general formula $\text{C}_{30}\text{H}_{62}\text{O}$, also known as
- melissyl alcohol or myricyl alcohol.
- It is found in plant cuticle waxes and in beeswax.
- The name cetyl derives from the whale oil (Latin: cetus) from which it was first isolated

- **Cetyl Alcohol** can be found in moisturizer, facial moisturizer, conditioner, antiaging, hair color, hair bleaching, facial cleanser. hand cream, shampoo, lipstick, eye cream.

1. True Waxes

Bees-wax is secreted by the honeybees that use it to form the combs. It

is a mixture of waxes

chief constituent is myricyl palmitate

(30C) (16C)

- **2. Spermaceti** is a wax that is most often found in the head cavities of the sperm whale.

Properties of TAG.

Triacylglycerols (TAGs)

- The simplest lipids constructed from fatty acids are the triacylglycerols,
- Also referred to as;
- triglycerides,
- fats, or neutral fats or storage lipids.
- Triacylglycerols are composed of three fatty acids in ester linkage with a single glycerol

Structure of triacylglycerols

- The three fatty acids esterified to a glycerol molecule are usually not of the same type
- The fatty acid on carbon 1 is typically saturated,
- Whereas that on carbon 2 is unsaturated, • and that on carbon 3 can be either
- Simple triglycerides:
 - Fatty acids connected to glycerol are of the same type
 - e.g., tripalmitin.
- Mixed triglycerides:
 - Fatty acids are of different types, e.g., • stearo-diolein and palmito-oleo-stearin.
- The main difference between fats and oils is for oils being liquid at room temperature, whereas, fats are solids.
- This is mainly due to presence of larger percentage of unsaturated fatty acids in oils than fats that has mostly saturated fatty acids. TAGs containing saturated fatty acids are solid at room temperature such as butter whereas
- TAGs containing unsaturated fatty acids are liquid at room temperature such as olive oil.(Olic acid, 18:1,9)

- What is lipoxins? 2

Which are more potent and have more defined physiological roles.

- The leukotrienes(LTs) and Lipoxins(LXs),

- Have 3 to 5 double bonds as described earlier, in comparison to prostaglandins which have 1 to 3 double bonds. The lipoxins are formed through the action of 15-lipoxygenase followed by the action of 5-lipoxygenase on arachidonic acid.

- A series of reductions of the resultant hydro-per-oxy groups leads to the

formation of tri-hydroxy derivatives of arachidonic acid known as the

lipoxins. Lipoxins induce chemotaxis and stimulate superoxide radicals for killing of microorganisms

- Prostaglandins, thromboxanes, leukotrienes and lipoxins have very short half

lives and rapidly degraded in the body.

- In summary, Eicosanoids are derived from C20 (eicosanoic) fatty acids

synthesized from the essential fatty acids and make up important groups of

physiologically active compounds.

Three examples of polyunsaturated fatty acid? 2

PUFAs The family of polyunsaturated fatty acids (PUFAs) with a double bond between

the third and fourth carbon from the methyl end of the chain are of special

importance in human nutrition

- **Arachidonic acid 20:4 Δ 5,8,11,14 is referred to as an ω -6 fatty acid because**

the closest double bond to the ω end begins six carbons from that end.

- **The essential fatty acid linoleic acid, 18:2 Δ 9, 12 is ω -6 fatty acid.**

- **In contrast, α -linolenic acid, 18:3 Δ 9, 12, 15 is an ω -3 fatty acid**

Isozyme of hexokinase illustrate the significance? 5

he first step in glucose metabolism in most cells, the transfer of a phosphate from ATP to glucose to form glucose 6-phosphate.

Hexokinase I, the isozyme in red blood cells has a low K_m for glucose of approximately

0.05 mM- helpful in utilizing blood glucose even when the blood glucose concentration is very low.

- **The isozyme of hexokinase, called glucokinase, which is found in the liver has a much**

higher K_m of approximately 5 to 6 mM- helpful in storing large amounts of “excess”

glucose as glycogen or converting it to fat after a carbohydrate meal.

46- Beta bends? 5

nomenclature of prostaglandins ? 5

Nomenclature of prostaglandins (PGs) involves the

- **assignment of a capital letter (PGE), • a numeral subscript (PGE1), and for the PGF family, a Greek letter subscript (PGF2 α).**

- **The capital letter refers to the ring substituents at positions X and Y.**

- **The numeral subscript that follows the capital letter(e.g. PGE1) refers to the PG series**

1, 2, or 3, determined by the number of unsaturated bonds present in the linear portion

of the hydrocarbon chain.

- **It does not include double bonds in the internal ring.**

- **The double bonds between carbons 13 and 14 are trans; the others are cis**

- **The Greek letter subscript, found only in the F series, refers to the position of the**

hydroxyl group at carbon 9.

- **This hydroxyl group primarily exists in the α position, i.e. it lies below the plane of the**

ring.

- **Prostaglandins have a hydroxyl group at C 15, a double bond between C 13 and C 14,**

and various substituents on 5 membered ring at C9 and C11

Double bonds also may

be present between carbons 5 and 6 and between carbons 17 and 18 in case of series

5 properties of waxes? 5

Waxes are insoluble in water, but soluble in fat solvents and are negative for acrolein test. very resistant to rancidity.

- **Waxes are not easily hydrolyzed as the fats and are indigestible by lipases**

(enzymes responsible for fat digestion in body) •

Thus they are of no nutritional value • Waxes

are of two types:

- **True waxes • Other Waxes or Non true waxes or**

- **Wax-like compounds**

- **True**

- **Waxes are solid simple lipids containing a monohydric alcohol (with a higher molecular**

weight than glyceric acid esterified to long-chain fatty acids.

- **Triacontanol is a fatty alcohol of the general formula $C_{30}H_{62}O$, also known as melissyl**

alcohol or myricyl alcohol.

- **It is found in plant cuticle waxes and in beeswax.**

- **Waxes are widely distributed in nature such as the secretion of certain insects as;**

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- **melissyl alcohol** or **myricyl alcohol**.
- It is found in plant cuticle waxes and in beeswax.
- The name **cetyl** derives from the whale oil (Latin: *cetus*) from which it was first isolated
- **Cetyl Alcohol** can be found in moisturizer, facial moisturizer, conditioner, antiaging, hair color, hair bleaching, facial cleanser. hand cream, shampoo, lipstick, eye cream

effect of pH on ionization of active site.

The concentration of H^+ affects reaction velocity in several ways.

First, the catalytic process usually requires that the enzyme and substrate have specific chemical groups in either an ionized or un-ionized state in order to interact.

For example, catalytic activity may require that an amino group of the enzyme be in the protonated form ($-NH_3^+$).

At alkaline pH, this group is deprotonated, and the rate of the reaction, therefore, declines.

Extremes of pH can also lead to denaturation of the enzyme.

.allosteric effect of hemoglobin.

Apart from Oxygen, ability of hemoglobin to reversibly bind oxygen is affected

by

- the pH of the environment**
- the $p\text{CO}_2$; and**
- the availability of 2,3-bisphospho-phoglycerate**
- These are collectively called Allosteric (other site) effectors**
- because their interaction at one site on the hemoglobin molecule affects the**

binding of oxygen to heme groups at other locations on the molecule

- In Greek, Allo means other and Steric means Site.**
- The binding of oxygen to myoglobin is not influenced by the allosteric effectors**

of hemoglobin

- In addition to transporting O_2 from the lungs to the peripheral tissues,**
- hemoglobin transports CO_2 , the by product of respiration,**
- And protons from peripheral tissues to the lungs**
- Transport of Carbon Dioxide**
- Hemoglobin carries CO_2 as carbamates formed with the amino terminal**

nitrogens of the polypeptide chains.

properties of nitrogenous base.

- **Aromatic:** The Nitrogen containing bases are aromatic i.e. they have alternate double bonds
- **Heterocyclic:**
 - They are heterocyclic i.e. structures that contain other atoms in addition to carbon, such as nitrogen in the ring structure
 - The six-atom rings of purines and pyrimidines are numbered in opposite directions.
 - **Weak Bases:** Purines or pyrimidines with an -NH_2 group are weak bases
 - **Functional Groups:** The most important functional groups of pyrimidines and purines are
 - ring nitrogens
 - carbonyl groups
 - exocyclic amino groups
 - **Hydrophobicity:** • The purine and pyrimidine bases are hydrophobic and relatively insoluble in water at the near-neutral cell pH
 - **Stacking Interaction:** Hydrophobic stacking interactions in which two or more bases are positioned with the planes of their rings parallel (like a stack of coins) are one of two important modes of interaction between bases in nucleic acids.

- Base stacking helps to minimize contact of the bases with water, and these interactions are very important in stabilizing the three-dimensional structure of nucleic acids.

define velocity' effect on the rate of velocity.

properties of glycerol.

Glycerol is widely used in pharmaceutical and cosmetic preparations.

- It has the following properties:
- Colorless Viscous oily liquid with sweet taste.
- Acrolein Test
- On heating with sulfuric acid or KHSO_4 (dehydration)
- it gives acrolein that has a bad odor.
- used for detection of free glycerol or any compound containing glycerol.
- In contrast to glycerol
- Sphingosine does not show positive acrolein test.
- Therefore glycerolipids and shingolipids can be differentiated on the basis of acrolein test.
- Glycerol combines with three molecules of nitric acid to form Glycerol trinitrate that is
- used as explosive and vasodilator
- On esterification with fatty acids it gives:
- monoacylglycerol: one fatty acid + glycerol.

- diacylglycerol: two fatty acids + glycerol.
- triacyl-glycerol: three fatty acids + glycerol.
- Gluconeogenic substrate Glycerol can form dihydroxyacetone phosphate which can convert into glucose or glyceraldehyde phosphate to enter into gluconeogenic pathway

carbon atoms are numbr in pentose sugar of Nucleotide and Nucleotide.

define kareitins

The protein keratin, formed by all vertebrates, is the chief structural component of hair, scales, horn, wool, nails, and feathers

Hemoglobin adult Mai kon c hoti hai.

Adenine guanine waly aye thy 2 mcq...

Water ka molecular weight kitna hota...

3.how activation energy effect rate of reaction

(3 functions of cGMP)

➤ serve as second messengers in signal transduction pathways.

Signal Transduction: GTP and GDP play

key roles in activating or inhibiting proteins in various cellular signaling cascades.

➤ Medical applications Specifically medical applications include the use of synthetic purine and

pyrimidine analogs that contain halogens, thiols, or additional nitrogen atoms;

➤ Their use includes chemotherapy for cancer

➤ as suppressors of the immune response during organ transplantation.

(reaction velocity in terms of order of reaction)

The rate or velocity of a reaction (v) is the number of substrate molecules converted to product per unit time;

Velocity is usually expressed as μmol of product formed per minute.

The rate of an enzyme-catalyzed reaction increases with substrate concentration until a maximal velocity

(V_{max}) is reached- reflecting the saturation with substrate of all available binding sites are occupied on the enzyme molecules present. (V_{max}) is reached- reflecting the saturation with substrate of all available binding sites are occupied on the enzyme molecules present

.(forces involved in three dimensional structure of proteins)

The Tertiary Structure

- Molecular forces that stabilise three dimensional tertiary structure include:
- Electrostatic Interactions
- Hydrophobic interactions
- Disulphide bridges

- **Hydrogen bonding**
- **The Tertiary Structure**
- **The positive charges on the basic amino acids enables them to form ionic**

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side chains of acidic amino acids.

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such as between aromatic groups of phenylalanine side chains.

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- **The Tertiary Structure Disulfide Bridges**

The amino acid cysteine in a protein can form a covalent disulfide bond

with another cysteine molecule through spontaneous (nonenzymatic)

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oxygen) in the peptide backbone and an oxygen atom in an amino acid

side chain.

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(non_competitive inhibitors in terms V_{max})

Noncompetitive Inhibition

- **Inhibitors bind enzymes at sites distinct from the substrate-binding site and**

- **generally bear little or no structural resemblance to the substrate.**

**Noncompetitive inhibitors bind at a separate site,
but may bind to either E or ES.**

- **Noncompetitive inhibition cannot be overcome by increasing the concentration of substrate.**

1. Effect on V_{max}

- **the apparent V_{max} changes, because the inhibitor is capable of preventing catalysis**

regardless of whether the substrate is bound to the enzyme.

- **Noncompetitive inhibition cannot be overcome by increasing the concentration of substrate.**

- **Thus, noncompetitive inhibitors decrease the V_{max} of the reaction.**

- **V_{max} decreases in noncompetitive inhibition.**

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- **The noncompetitive inhibitor, in effect, lowers the concentration of the active enzyme**

and therefore decreases the V_{max} of the enzyme.

Saponification/Saponification number

- o **Hydrolysis of a fat by an alkali is called saponification**

- o **The resultant products are;**

o glycerol and the alkali salts of the fatty acids, which are called “soaps”

o The number of mgs of NaOH/KOH required to saponify the free and combined FA in one gram of a given fat is called its saponification number.

The amount of alkali needed to saponify a given quantity of fat will depend upon the number of carboxylic ($-\text{COOH}$) group present .

Thus fats containing short chain fatty acids will have more $-\text{COOH}$ groups per gram than long- chain

fatty acids and this will take up more alkali And hence will have higher saponification number

3 Induced fit in protein synthesis.

4 Steróls and name of steròle which is present in vertebrates.

which two sugars are present in nucleic acid write characteristics.

Pentose Sugar

- D-ribose and 2-deoxy D-ribose are the only sugars so far found in the nucleic acids.

- These aldo pentoses belong to D-family

- They are present as Furanose (ring) in the form of β -Anomer

The addition of a pentose sugar to a base produces a nucleoside

- If the sugar is D-ribose, a ribonucleoside is produced

- If the sugar is 2-deoxy D- ribose, a deoxyribonucleoside is produced

enzyme rate of substration

types of RNA

Messenger RNA (mRNA)

This class is the most heterogeneous in

- o Abundance Size (500-6000 nucleotides)**

- o base sequence**

- o Stability**

- o mRNA comprise about 2–5% of total cellular RNA**

- o mRNA molecules are formed with the help of DNA template strand (3′-5′) during the process called transcription.**

- o In addition to the protein coding regions in the mature eukaryotic mRNA that can be translated,**

- o there are untranslated regions at its 5′ and 3′ ends**

- o Moreover, there is a 5′ cap and a poly A tail at 3′ end**

- o Function of mRNA**

- o The members of this class function as messengers to convey the information in a gene to the protein synthesizing machinery.**

- o The mRNA carries genetic information from the nuclear DNA to the cytosol, where it is used as a**

template for protein synthesis.

71. Transfer RNA (tRNA)

- o t RNA is the smallest of the three major species of RNA (4S).**
- o They are single stranded globular molecules.**
- o They remain largely in cytoplasm.**
- o They are generated by nuclear processing of a precursor molecule.**
- o tRNAs compose roughly 20% of total cellular RNA There are at least 20 species of tRNA molecules in every cell.**
- o Although each specific tRNA differs from the others in its sequence of nucleotides, the tRNA molecules as a class have many features in common**
 - o Primary structure**
 - o t RNA molecules consist of 74-95 nucleotides in a particular sequence.**
 - o The t RNA molecules contain not only the usual bases like adenine, guanine, cytosine, uracil but also contain unusual bases**
 - o These unusual bases(also called modified bases) include**
 - o Dihydrouracil Pseudouridine**
 - o Arms or loops of tRNA**
 - o All tRNA molecules contain 4 main arms or loops.**

- o **1-Acceptor arm:** This is made up of unpaired sequences of cytosine-cytosineadenine (CCA) at the 3' end.
- o **The 3'OH group of adenine binds with the carboxylic group of a specific amino acid and carries it to ribosomes for protein synthesis.**
- o **2-Anticodon arm:** It is in the form of a loop and carries specific sequences of three bases which constitute the anticodon.
- o **The bases of anticodon are bonded with three complementary bases of codon on mRNA**
- o **Function of tRNA**
- o **The t RNA molecules serve as ADAPTERS for the translation of information in the sequence of nucleotides of the mRNA into specific amino acids.**
- o **There is at least one (and often several) specific type of tRNA molecule for each of the amino acids commonly found in proteins.**
- o **Each t RNA carries its specific amino acid to the site of protein synthesis.**
- o **There it recognizes the genetic code word on mRNA (codon) and this specifies the addition of its amino acids to the growing peptide chain.**

72. Ribosomal RNA (rRNA)

- o **found in association with several proteins as a component of ribosomes---** a cytoplasmic nucleoprotein

structure that acts as the machinery for the synthesis of proteins from the mRNA template.

- o RNAs make up 80% of the total RNA in the cell.**

- o The ribosomal subunits are defined according to their sedimentation velocity in Svedberg units.**

- o Svedberg unit is related to the molecular weight and shape of the compound.**

- o The bases in r RNA are mainly adenine, guanine, cytosine and uracil and a few pseudouridine**

- o Eukaryotic Ribosome**

- o The mammalian ribosome contains two major nucleoprotein subunits:**

- o a larger one with 60S a smaller one with 40S.**

- o The 60S subunit contains**

- o a 5S rRNA**

- o a 5.8S rRNA**

- o a 28S rRNA**

- o more than 50 specific polypeptides.**

- o The 40S subunit is smaller and contains a single 18S rRNA**

- o Approx. 30 distinct polypeptide chains.**

- o In eukaryotes, all of the ribosomal RNA molecules except the 5S rRNA, which is independently**

transcribed, are processed from a single 45S precursor RNA molecule in the nucleolus- packed with the

specific ribosomal proteins.

- o The rRNA are necessary for ribosomal assembly and play a key role in the binding of mRNA to**

ribosomes and its translation

- o In the cytoplasm, the ribosomes remain quite stable and capable of many translation Cycles**

73. Other types of RNA Small nuclear RNA (snRNA)

Small nuclear RNA (snRNA) are large number of small stable RNA species found in eukaryotic cells.

- o Most of them are complexed with proteins to form ribonucleoproteins. They are distributed in the nucleus, in the cytoplasm or in both.**

- o They are significantly involved in rRNA and mRNA processing and gene regulation.**

- o Large & Small Noncoding Regulatory RNAs**

- o One of the most exciting discoveries in the last decade of eukaryotic regulatory biology has been the**

identification and characterization of regulatory nonprotein coding RNAs (ncRNAs).

- o NcRNAs exist in two general size classes,**

- o small consisting of microRNA (miRNAs) and silencing (siRNAs) and**

- o Large consisting of long noncoding RNAs (lncRNAs)**

.:properties of D-fructose

cataracts D-Fructose:

Commonly known as fruit sugar as it occurs free in fruits, honey etc

Also occurs as a constituent of sucrose

It is levorotatory and hence is also called Levulose

Seminal fluid is rich in fructose and sperm utilize fructose for energy

Readily metabolized either via glucose or directly

Hereditary fructose intolerance leads to fructose accumulation

Platelet-activating factor (PAF)

This is an unusual ether glycerophospholipid, with a saturated alkyl group in an ether link to carbon 1

and an acetyl residue (rather than a fatty acid) at carbon 2 of the glycerol backbone.

PAF is synthesized and released by a variety of cell types

It binds to surface receptors, triggering potent thrombotic and acute inflammatory events .

It causes platelets to aggregate and degranulate (required for clotting), and neutrophils and alveolar

macrophages to generate superoxide radicals (required for microbial killing)

rate of order of reaction

Order of Reaction

When $[S]$ is much less, then the velocity of the reaction is approximately proportional to the substrate concentration.

The rate of reaction is then said to be first order with respect to substrate.

When $[S]$ is much greater than K_m the velocity is constant and equal to V_{max} .

The rate of reaction is then independent of substrate concentration, and is said to be zero order with respect to substrate concentration

Calculate the pK_a of lactic acid, given that when the concentration of lactic acid is 0.01M and the concentration of lactate is 0.087 M, the pH is 4.80

- $pH = pK_a + \log [lactate]/[lactic\ acid]$
- $pK_a = pH - \log [lactate]/[lactic\ acid]$

$$= 4.80 - \log 0.087/0.01$$
$$= 4.80 - \log 8.7$$
$$= 4.80 - 0.94$$
$$= 3.9$$

name three pathway for the Arachdeonic acid

Arachidonic acid is the most common precursor of the eicosanoids. Three major pathways for the metabolism of arachidonic have been discovered so far.

- ❖ Cyclooxygenase pathway; prostaglandins and thromboxanes.
- ❖ lipoxygenase pathway; leukotrienes, HETEs, and lipoxins.
- ❖ cytochrome P450 pathway epoxides and HETEs
- ❖ There are different series of eicosanoids
- ❖ depending on the precursor 20 C FA. derived from the essential fatty acids linoleic acid and linolenic,
OR directly from dietary arachidonic acid and eicosapentaenoic acid
- ❖ Depending on the precursor different numbers of double bonds are present in these eicosanoids reflecting the parent 20C FA.

name the hydrolases

Hydrolases

These enzymes catalyze hydrolysis, i.e.

o add water molecule to the substrate which is simultaneously decomposed; the functional group of substrate is transferred to water.

o Common example of hydrolases are:

o Protein hydrolyzing Enzymes (peptidases).

o Carbohydrases

o Lipid hydrolyzing enzymes e.g. Lipases and

❖ Lyases

These enzymes catalyze the addition of NH_3 , H_2 O or CO_2 to double bonds or the removal of these groups leaving behind double bonds. Lyases are included in a separate class because they catalyze these reactions by means other than hydrolysis or oxidation.

❖ Isomerases

These enzymes catalyze the structural change within a single molecule by the transfer of groups within it, resulting in the formation of an isomeric form of substrate.

❖ Ligases

These enzymes catalyze condensation reactions joining two molecules by forming

- o C-O,**
- o C-S,**
- o C-N and**
- o C-C bonds.**
- o The energy for condensation is provided by cleavage of high energy phosphates, e.g. ATP, GTP etc.**

number of carbon in nucleotide and nucleoside

The carbon and nitrogen atoms in the rings of the base and the sugar are numbered separately

The atoms in the rings of the bases are numbered

1 to 6 in pyrimidines &

1 to 9 in purines

In the pentoses of nucleotides and nucleosides the carbon numbers are given a prime (') designation to distinguish them from the numbered atoms of the nitrogenous base.

The carbons in the pentose are numbered 1' to 5'.

Numerals with a prime (e.g., 2' or 3') distinguish atoms of the sugar from those of the heterocycle

:disulphide bridge

Disulfide bridges are sometimes called disulfide bonds or S-S bonds. They are covalent links between

the Sulphur atoms of two cysteine amino acids and their formation stabilizes the tertiary and higher order

what do you know about km of michaelis- menton equation (3marks)

Michaelis-Menten equation, the rate equation (Contd.)

$$V_0 = \frac{V_{\max} [S]}{K_m + [S]}$$

$$K_m + [S]$$

- Where;
- V_0 = initial reaction velocity
- $V_{\max}$ = maximal velocity
- K_m = Michaelis constant

- $= \frac{K_1 + K_2}{K_1}$
- $[S]$ = substrate concentration
- This is the Michaelis-Menten equation-statement of the quantitative relationship between the;
 - initial velocity V_0 ,
 - the maximum velocity V_{max} , and
 - the initial substrate concentration $[S]$, all related through the Michaelis constant K_m
- An important numerical relationship emerges from the Michaelis-Menten equation in the special case;
 - when V_0 is exactly one-half V_{max}
$$V_0 = \frac{V_{max} [S]}{K_m + [S]}$$

$$\frac{1}{2} V_{max} = \frac{V_{max} [S]}{K_m + [S]}$$
- On dividing by V_{max} , we obtain
- $\frac{1}{2} = \frac{[S]}{K_m + [S]}$
- Solving for K_m , we get
- $K_m + [S] = 2[S]$, or
- $K_m = [S]$, when $V_0 = \frac{1}{2} V_{max}$

- This is a very useful, practical definition of K_m : K_m is equivalent to the substrate

concentration at which V_0 is one-half V_{max} .

- The equation describes the kinetic behavior of a great many enzymes, and all enzymes

that exhibit a hyperbolic dependence of V_0 on $[S]$ are said to follow Michaelis- Menten

kinetics.

- The practical rule that $K_m = [S]$ when $V_0 = 1/2 V_{max}$ holds for all enzymes that follow

Michaelis-Menten kinetics.

Q6:-what do you know about N-glycosidic. And how it is formed(5marks)

N-glycosidic bond

- Sugars are linked to the heterocycle by a β -N-glycosidic bond, almost always to the

- N-1 of a pyrimidine

- N-9 of a purine

- The N-glycosyl bond is formed by removal of the elements of water

- a hydroxyl group from the pentose and

- hydrogen from the base

Thus it is a condensation reaction.

- Similar to O-glycosidic bond formation in carbohydrates
- However, N-glycosidic bonds, have Nitrogen atom instead of oxygen linking the two residues.
- the addition of the glycosidic bond to nitrogenous base is indicated by the name change
- such as from adenine to adenosine for the glycosidic bond

Name of hypercholesterolemic (3mark)

Lauric (12:0), myristic (14:0), and palmitic (16:0) acids are considered to be Hypercholesterolemic

Velocity link with enzymes concentration

Relationship of velocity to enzyme concentration

- The rate of the reaction is directly proportional to the enzyme concentration.
- There is a linear relationship between reaction rate and enzyme concentration (at constant substrate concentration)
- For example,
- if the enzyme concentration is halved, the initial rate of the reaction (V_0) as well as that of V_{max} are reduced to one half that of the original

Hydrolytic or oxyletic rancidity

Oxidative Rancidity

- Oxidation of the fat molecules give rise to some short • chain aldehydes, ketones and dicarboxylic acids
- which have objectionable taste and odor.
- The unpleasant taste and smell associated with rancidity result from the oxidative cleavage of double bonds in unsaturated fatty acids
- The oxygen of the air is necessary for this type of • rancidity. This can be prevented by addition of anti-oxidants such as vitamin E to foods.
- II. Hydrolytic Rancidity
- It is due to the slow hydrolysis of fats,
- which in case of fats like butter results in the liberation of short chain fatty acids which are volatile and have rancid taste and odor.

Nucleotide role as energy source in cell

Energy currency: Nucleotides play an important role as "energy currency" in the cell.

- Nucleoside tri- and diphosphates such as ATP and ADP are the principal donors and acceptors of phosphoryl group in metabolism.
- By doing this, they play a key role in the energy transduction.

- This energy is used in almost every energy requiring process in the body, such as;

- Muscle contraction, Transmission of nerve impulse, Transports of nutrients across cell

membrane Motility of spermatozoa And many more energy dependent processes

Beta bends

Beta-bends are short regions usually involving four successive amino acid Residues.

- They often connect strands of antiparallel beta-sheets.

- β -Bends reverse the direction of a polypeptide chain, helping it form a

Compact, globular shape.

- The structure is a 180 degree turn involving four amino acid residues

- The carbonyl oxygen of the first residue forms a hydrogen bond with the Amino-group hydrogen of the fourth residue.

- β -Bends were given this name because they often connect successive Strands of antiparallel β -sheets

- Glycine and Proline residues often occur in β turns

- Glycine because it is small and fl β -sheet

- Proline because peptide bonds involving the imino nitrogen readily Assume the cis configuration, a form that is particularly amenable to a tight Turn

name Nucleotide components

Pentose sugar

Bases

Phosphate group

Name 3 pathways of Arachidonic metabolism

Cyclooxygenase pathway; prostaglandins and thromboxanes.

- **lipoxygenase pathway; leukotrienes, HETEs, and lipoxins.**
- **cytochrome P450 pathway epoxides and HETEs**

what is nucleotide “regulatory role”

: Nucleotides are important regulatory compounds for many of the pathways of intermediary metabolism, inhibiting or activating key enzymes.

- **Roles that nucleotides perform in metabolic regulation include:**
- **ATP-dependent enzyme phosphorylation in key metabolic reactions.**
- **Allosteric regulation of enzymes by ATP, AMP, and CTP**
- **Phosphorylation can either activate or inhibit ATP- dependent enzyme depending on the type of enzyme.**
- **Similar is the case with dephosphorylation**

Induced fit in protein synthesis.

Some proteins can change their shape (conformation)

- **When a substrate combines with an enzyme, it induces a change in the**

enzyme's conformation

- This change in conformation when the substrate binds is induced by multiple

weak interactions with the substrate. • There may also be rearrangements of

covalent bonds during an enzyme-catalyzed reaction.

- This conformational change is referred to as induced fit. Chemical reactions of many types take place between substrates and

enzyme's functional groups (specific amino acid side chains, metal ions, and coenzymes)

- Induced fit serves to bring specific functional groups on the enzyme into the proper position to catalyze the reaction.

Sterols and name of sterol which is present in vertebrates.

Steroids with eight to ten carbon atoms in the side chain at C-17 and a hydroxyl group at C-3

are classified as sterols

- Cholesterol is the major sterol in vertebrates

- Cholesterol has an eight carbon branched hydrocarbon chain attached to C-17 of the D ring •

Ring A has a hydroxyl group at C-3, and ring B has a double bond between C-5 and C-6

What is the concentration of H^+ in a solution of 0.1 M NaOH?

Solution: $K_w =$

$[H^+][OH^-]$

❖ With $[OH^-] = 0.1 \text{ M}$, solving for $[H^+]$ gives

❖ $[H^+] = K_w/[OH^-] = 1 \times 10^{-14} \text{ M}^2 / 0.1 \text{ M} = 10^{-14} \text{ M}^2 / 0.1 \text{ M} = 10^{-13} \text{ M}$

which two sugars are present in nucleic acid write characteristics.

Pentose Sugar

- D-ribose and 2-deoxy D-ribose are the only sugars so far found in the nucleic acids.
- These aldo pentoses belong to D-family
- They are present as Furanose (ring) in the form of β -Anomer

The addition of a pentose sugar to a base produces a nucleoside

- If the sugar is D-ribose, a ribonucleoside is produced
- If the sugar is 2-deoxy D-ribose, a deoxyribonucleoside is produced

enzyme rate of substration

Lactose characteristics

Lactose, the disaccharide of milk consists of galactose joined to glucose by

β (1 → 4) glycosidic linkage

- It is dextrorotatory
- As one of the aldehyde group is free, it has reducing properties and exhibits

mutarotation

- In lactating mammary gland, the lactose is synthesized from glucose by the duct epithelium
- Many organisms that are found in milk, e.g., E.Coli convert lactose of milk to lactic acid (by β galactosidase) thus causing souring of milk
- Lactase, the specific enzyme which hydrolyses lactose is present in the intestinal brush border

PI value of glutamate

The pI of glutamate is ____ 3.22

Classification of protein

Different classifications of proteins are based on their:

1. Shape and Size
2. Biological actions/ Functions
3. Solubility and physical properties
4. Quality Catalytic Proteins:

These specialized proteins are called enzymes which catalyze the biochemical reactions

2. Protective Proteins

a) Immunoglobins (Igs)

- **These freely circulating proteins protect the body from invading microbes such as bacteria or viruses by inactivating or killing them through**

various mechanisms. b) Fibrinogen

- **This forms fibrin clot and stops bleeding from wounds**

3. Regulatory Proteins:

- **Hormones control genetic expression, cellular signalling and biochemical**

reactions catalyzed by enzymes. Enzymes are either activated or inactivated through modification of their structure.

- **Examples of protein hormones are**

- **growth hormone**
- **Insulin**
- **Glucagon – Somatostatin**

4. Structural Proteins:

- **These proteins form various body structures e.g.**

- **Collagen**
- **Elastin**
- **Keratin**

5. Transport Proteins:

- **These proteins transport various substances from one part of the body to**

the other e.g.

– Hemoglobin transport O₂ from lungs to tissues and CO₂ from tissues to lungs – Transferrin transports iron

6. Contractile Proteins:

- These proteins are involved in muscle contraction and relaxation**

- Myosin of thick filaments**

- Actin of thin filaments of skeletal muscles**

7. Respiratory Proteins:

- Heme containing proteins are involved in the function of respiration e.g.**

Hemoglobin

Myoglobin

cytochromes

8. Digestive Proteins:

- These proteins are digestive enzymes which digest our food materials such as**

carbohydrates, proteins, lipids and include

Amylase

Pepsin

Lipases etc.

9. Toxin Proteins:

- These proteins are hydrolytic enzymes found in the venom of poisonous snakes, sting**

of bees and insects and hydrolyze the compounds forming the structure of the cell

membrane

9. Toxin Proteins:

- These proteins are hydrolytic enzymes found in the venom of poisonous snakes, sting of bees and insects and hydrolyze the compounds forming the structure of the cell membrane The poisonous mushrooms also have such toxins

10. Storage Proteins:

- These proteins store some specific elements or compounds with them

- This is because of the presence of the many binding sites in them for the

particular element e.g.

- ferritin stores iron
- ceruloplasmin stores copper

Lactose intolerance?

Lactose intolerance refers to a condition of pain, nausea, and flatulence

after the ingestion of foods containing lactose, most notably dairy products

- Although it is often caused by
 - low levels of lactase
 - it also can because by intestinal injury

- In a lactase deficient person, lactose accumulates in the lumen of the

small intestine after ingestion of milk

- The large osmotic effect of the unabsorbed lactose leads to an influx of fluid

into the small intestine

- Hence, the clinical symptoms of lactose intolerance are:

- abdominal distention,

- nausea,

- cramping,

- pain

- a watery diarrhea

Midertanean diet?

The Mediterranean diet contains

- seasonally fresh food, with

- an abundance of plant material,

- low amounts of red meat, and

- olive oil as the principal source of fat

- The Mediterranean diet is associated with decreased serum total cholesterol

- and LDL cholesterol—but

- little change in HDL cholesterol—when compared with a typical Western diet higher in

saturated fats

Write types of lipoprotein?

- The lipoprotein particles include
 - chylomicrons (CM),
 - very-low-density lipoproteins (VLDL), • low-density lipoproteins (LDL), and
 - high-density lipoproteins (HDL).
 - They differ in • lipid and protein composition,
 - size, • density and • site of origin
 - Different combinations of lipids and proteins produce particles of different densities
 - ranging from chylomicrons to high-density lipoproteins
 - these particles can be separated by ultracentrifugation increase
 - Chylomicrons are formed and assembled in intestinal mucosal cells after a fatty meal
 - And carry dietary triacylglycerol, cholesterol, and cholesteryl esters to the peripheral
- tissues. • As the chylomicron circulates and most of its dietary TAG are degraded and
- taken up by peripheral tissues in the form of fatty acids,
- the particle size decreases and density increases.
 - The remaining particle, called Chylomicron Remnant, is removed from the circulation by
- the liver.

- VLDLs are assembled in the liver.
 - composed predominantly of TAGs synthesized in liver and
 - contain some cholesterol and cholesteryl esters
 - As VLDL pass through the circulation, TAG is degraded and taken up by peripheral tissues in the form of fatty acids,
 - causing the VLDL to decrease in size and become denser,
 - called VLDL remnants
- further removal of
- from these remnants produces low-density lipoprotein (LDL)
 - LDL particles contain much less triacylglycerol than their VLDL predecessors,
 - and have a high concentration of cholesterol and cholesteryl esters
 - LDLs contain apoB-100 as their major apolipoprotein
 - they carry cholesterol to extra hepatic tissues that have specific plasma membrane receptors that recognize apoB-100.
 - apoB-100 receptors internalize the LDL into the cell.
 - Remaining LDL is endocytosed by Liver cells.
 - Oxidized LDL can also accumulate in the macrophage cells lining the arteries
 - resulting in the formation of atherosclerosis.

HDL The fourth major lipoprotein type, high-density lipoprotein,

- **originates in the liver and small intestine**
- **as small, protein-rich particles that contain little cholesterol and no cholesteryl esters**
- **They take up cholesterol from non-hepatic (peripheral) tissues and return it to the liver as cholesteryl esters**
- **When cholesterol is taken up by HDL, it is immediately esterified and becomes hydrophobic and**
- **which is sequestered in the core of the HDL,**
- **This transforms the nascent HDL to a mature HDL particle.**
- **Mature HDL then returns to the liver, where the cholesterol is unloaded in a process called reverse cholesterol transport pathway**
-
- **This is the basis for the inverse relationship seen between plasma HDL concentration and atherosclerosis, and for HDL's designation as the "good" cholesterol carrier.**
- **A much stronger correlation exists between the levels of blood LDL cholesterol and heart disease**

- In contrast, high levels of HDL cholesterol have been associated with a decreased risk for

heart disease

- Abnormal levels of plasma lipids (dyslipidemias) act in combination with smoking, obesity,

sedentary lifestyle, insulin resistance, and other risk factors to increase the risk of CHD

- Clinical studies have demonstrated that dietary or drug treatment of hyper -

cholesterolemia is effective in decreasing LDL, increasing HDL, and reducing the risk for

cardiovascular events

Starch structure?

Starch contains two types of glucose polymers:

- Amylose

- amylopectin

- It is the most important dietary carbohydrate in cereals, potatoes, legumes,

and other vegetables

- Amylose (13–20%), consists of long, unbranched helical structure of D-

glucose residues connected by ($\alpha 1 \rightarrow 4$) linkages (as in maltose)

Hydrogenation?

Converts Unsaturated fatty acids to Saturated fatty acids.
Margarines are vegetable oils treated with partial hydrogenation to form semi- solid consistency

‘butter’

o Hydrogenated fats are used by many commercial food producers to Provide rich texture , Increase shelf

live , Increase melting point .

o Resistance to oxidation and flavour deterioration

o Partial hydrogenation has an undesirable, effect: some cis double bonds are converted to trans double bonds.

o There is a strong evidence that dietary intake of trans fatty acids

o (often referred to as "trans fats") leads to a higher incidence of cardiovascular disease, therefore these

fats should be avoided in the diet.

name 3 bases of pyrimidine

Pyrimidines include: • Cytosine (C)—in both DNA and RNA

• Thymine (T)—only in DNA

• Uracil (U) —only in RNA

Cytosine when combines with pentose it becomes deoxycytidine and cytidine

• Thymine becomes thymidine and deoxythymidine

• Uracil (U) becomes uridine and deoxyuridine depending on the type of sugar.

- Cytosine is 2-oxy-4-amino-pyrimidine
- Thymine is 2,4-dioxy-5-methyl-pyrimidine • Uracil is 2,4-dioxy-pyrimidine
- T and U differ by only one methyl group, which is present on T but absent on U
- Minor Pyrimidine Bases: • Dihydrouridine (DHU) ,
- 5-Methyl Cytadine & • 5-Hydroxy-Methyl Cytadine
- Unnatural Pyrimidine Bases:
- Fluorouracil (5FU) &
- 6-Aza Cytosine (AZC)

What do you know about Km of kinetic equation

rancidity, factors contribute in it

The chemical deterioration of fats.

When lipid-rich foods are exposed too long to the oxygen in air, they may spoil and become foul smelling.

It is a physico-chemical change in the natural properties of the fat leading to the development of unpleasant odor or taste or abnormal color .

It occurs particularly on aging after exposure to atmospheric oxygen, light, moisture, bacterial or fungal contamination and/or heat.

Saturated fats resist rancidity more than unsaturated fats that have unsaturated double bonds.

B conformation in proteins

The β Conformation organizes polypeptide Chains in β sheets

β Sheets are a second type of regular secondary structure that maximizes

hydrogen bonding between the peptide backbones.

The backbone of the polypeptide chain is extended into a zigzag rather

than helical structure

The zigzag polypeptide chains can be arranged side by side to form a

structure resembling a series of pleats

In this arrangement called a β sheet, hydrogen bonds form between

adjacent segments of polypeptide chain

The β -pleated sheet is described as parallel if the polypeptide strands run in

the same direction (as defined by their amino and carboxy terminals) and

anti-parallel if they run in opposite direction

Antiparallel strands are often the same polypeptide chain folded back

on itself, with simple hairpin turns or long runs of polypeptide chain

connecting the strands The amino acid side chains(R groups) of each polypeptide strand alternate between extending above and below the plane of the β -sheet. Most sheets are not perfectly flat but

tend to have a twist.

The individual segments that form a β sheet are usually nearby on the

polypeptide chain, but can also be quite distant from each other in the

linear sequence of the peptide and may even be in different polypeptide

chains.

Explain km equation in detail Small and large km value

Small km: a low concentration of substrate is needed to half-saturate the enzyme—that is, to reach a velocity that is $1/2 V_{max}$

Large Km: A numerically large (high) Km reflects a low affinity of enzyme for substrate because a high concentration of substrate is needed to half saturate the enzyme

The velocity of an enzyme is most sensitive to changes in substrate concentration over a concentration range below its Km.

d. At substrate concentrations less than 1/10th of the K_m , a doubling

of substrate concentration nearly doubles the velocity of the reaction . At substrate concentration has little effect on velocity.

velocity effect on reaction rate

Characteristics of cyclic AMP

Cyclic AMP is a cyclic nucleotide

• cAMP is synthesized in tissues from ATP • chemically it is 3'-5' adenosine

monophosphate.

Functions of c-AMP

- Acts as second messenger in the cell**
- It has role in glycogen metabolism**
- cAMP, glycogenolysis**
- cAMP TAG metabolism**
- cAMP lipolysis**
- It decreases cholesterol synthesis**
- It causes activation of protein kinases which in turn;**
- activate or deactivate other enzymes.**
- It regulates the cell membrane permeability, by increasing permeability of cell**

membrane

- to H₂O, Na⁺, K⁺ & Ca²⁺
- Moreover, it regulates
- insulin secretion,
- catecholamine biosynthesis & Melatonin synthesis
- Cyclic GMP is synthesized from GTP
- It serves as a second messenger in response to nitric oxide during relaxation of smooth

muscle (especially blood vessels) so it has role in smooth muscle relaxation and

vasodilatation.

- It also has role in
- Protein phosphorylation
- Neurotransmission
- Insulin action
- Regulation of sodium channels

Zero order reaction

The rate of reaction is then independent of substrate concentration, and is said to be zero order with respect to substrate concentration

Rate = k

What is pka

Equilibrium constants for ionization reactions are usually called ionization

constants or acid dissociation constants, often designated pKa

- The stronger the tendency to dissociate a proton, the stronger is the

acid and the lower its pKa

- The pKa expresses, on a logarithmic scale, the relative strength of a weak

acid or base:

- $$\text{pH} = \text{pK}_a + \log \frac{[\text{A}^-]}{[\text{HA}]}$$
- $$\text{pK}_a = \log \frac{1}{K_a} = -\log K_a$$

- The stronger the acid, the lower its pKa

Medical use of nucleotide and nucleoside

Medical applications Specifically medical applications include the use of synthetic

purine and pyrimidine analogs that contain halogens, thiols, or additional nitrogen

atoms;

- Their use includes chemotherapy for cancer

- as suppressors of the immune response during organ transplantation.
- as anti-viral drugs such as in the treatment of AIDS

Role of Triacylglycerol in plant

Triacylglycerols are also stored as oils in the seeds of many types of plants.

- Thus providing energy and biosynthetic precursors during seed germination

Cyclooxygenase pathway

Prostaglandins are fatty acids containing 20 carbon atoms, including • an internal 5-carbon ring.

- Prostaglandins have
- a hydroxyl group at C 15,
- a double bond between C 13 and C 14, and
- various substituents on 5 membered ring at C9 and C11
-
- Nomenclature of prostaglandins (PGs) involves the
- assignment of a capital letter (PGE), • a numeral subscript (PGE1), and
- for the PGF family, a Greek letter subscript (PGF2 α).
- The capital letter refers to the ring substituents at positions X and Y.
-

- The numeral subscript that follows the capital letter(e.g. PGE1) refers to the PG series

1, 2, or 3, determined by the number of unsaturated bonds present in the linear portion of the hydrocarbon chain.

-

- It does not include double bonds in the internal ring.

- The double bonds between carbons 13 and 14 are trans; the others are cis

- The Greek letter subscript, found only in the F series, refers to the position of the

hydroxyl group at carbon 9.

- This hydroxyl group primarily exists in the α position, i.e. it lies below the plane of the

ring.

- Prostaglandins have a hydroxyl group at C 15, a double bond between C 13 and C 14,

and various substituents on 5 membered ring at C9 and C11

Double bonds also may

be present between carbons 5 and 6 and between carbons 17 and 18 in case of series

2 and 3

- Thromboxanes (TX)

- also formed via the cyclooxygenase pathway,

- differ from PGs in that they contain a 6-membered ring that includes an oxygen atom

-

- Biosynthesis of the PGS and TX

- Those derived from arachidonic acid, the 2-series, (such as PGE₂, TXA₂), are described

here • because the 1-series and the 3-series are present in very small amounts in

humans.

- The initial step, which is catalyzed by a cyclooxygenase (COX), forms the

fivemembered ring and

- adds four atoms of oxygen (two between C 9 and C11, and two at C 15) to

- form the unstable PGG₂.

- The hydro-peroxy group at carbon 15 is

- quickly reduced to a hydroxyl group by a peroxidase to form PGH₂ • PGH₂ is the

precursor of all other PGs and TXAs

- The next step is tissue specific

- For example, PGH₂ may be reduced to PGE₂ or PGD₂ by specific isomerases

(PGE synthase or PGD synthase respectively)

- . • For example, TXA Synthase is present in high concentration in platelets and forms

TXA₂.

- In the vascular endothelium, however, PGH₂ is converted to the PGI₂ by action of

Prostacyclin Synthase.

- The beneficial effect of cold water fish (e.g., salmon), with a high content of eicosa-

pentaenoic(TXA₃) acid (EPA), and docosa-hexaenoic acid (DHA) comes from the fact

that they lead to formation of more TXA₃ relative to TXA₂.

- TXA₃ is less effective in stimulating platelet aggregation than its counterpart in the 2-

series, TXA₂.

- Platelet aggregation is the culminating step in the cardiovascular diseases due to

atherosclerosis.

biomedical importance of oligosaccharides

Integral membrane proteins contain covalently attached carbohydrate

units, oligosaccharides, on their extracellular face

- Many secreted proteins such as antibodies and coagulation factors also

contain oligosaccharide units

- The oligosaccharides participate in molecular targeting and cell to cell recognition

They also mark the passage of time and determine when the proteins

should be taken out of circulation

- The human ABO blood groups illustrate the effects of glycosyl-transferases
- Carbohydrates are attached to glycoproteins and glycolipids on the surfaces of red blood cells
- For one type of blood group, one of the three different structures, termed A, B, and O, may be present
- These structures have in common an oligosaccharide foundation called the O (or sometimes H) antigen
- The A and B antigens differ from the O antigen by the addition of one extra

monosaccharide, either N-acetylgalactosamine (for A) or galactose (for B)

through an $\alpha 1,3$ linkage to a galactose moiety of the O antigen.

four structure of protein

Four levels of protein structure are commonly defined

The Primary Structure

A description of all covalent bonds (mainly peptide bonds and disulfide bonds)

linking amino acid residues in a polypeptide chain is its primary structure

The most important element of primary structure is the sequence of amino acid

residues Secondary structure

Secondary structure refers to particularly stable arrangements of amino acid

residues giving rise to recurring structural patterns

Tertiary structure

Tertiary structure describes all aspects of the three-dimensional folding of a

polypeptide

Quaternary structure

When a protein has two or more polypeptide subunits, their arrangement in

space is referred to as quaternary structure

phosphoethanolamine

The Sphingomyelins contain phosphocholine or phosphoethanolamines as their polar head group and are therefore classified along with glycerophospholipids as phospholipids

Glucogenic amino acids

- Amino acids whose catabolism yields pyruvate are termed glucogenic or glycogenic
- These intermediates are substrates for gluconeogenesis and, therefore, can give rise to the net formation of glucose or glycogen in the liver and glycogen in the muscle

	Glucogenic	Glucogenic And Ketogenic	Ketogenic
Nonessential	Alanine Arginine Asparagine Cysteine Glutamate Glutamine Glycine Proline Serine	Tyrosine	
Essential	Histidine Methionine Valine	Threonine Isoleucine Phenyl-alanine Tryptophan	Leucine Lysine

Calculate the ratio of the concentrations of acetate and acetic acid required in a

buffer system of pH 5.30 (pKa = 4.76)

$$\bullet \text{ pH} = \text{pKa} + \log [\text{acetate}]/[\text{acetic acid}] \bullet \log [\text{acetate}]/[\text{acetic acid}] = \text{pH} - \text{pKa}$$

$$= 5.30 - 4.76$$

$$= 0.54$$

$$\bullet [\text{acetate}]/[\text{acetic acid}] = \text{antilog } 0.54$$

$$= 3.5$$

Titration curve of glycine

From the titration curve of glycine we can derive several important pieces

of information

- First, it gives a quantitative measure of the pKa, of each of the two

ionizing groups:

- 2.34 for the -COOH group

- 9.60 for the -NH₃⁺ group

- The second piece of information is that this amino acid has two regions

of buffering power

- One of these is the relatively flat portion of the curve, extending for

approximately 1 pH unit on either side of the first pKa, of 2.6, indicating

that glycine is a good buffer near this pH

- **The other buffering zone is centered around pH 9.60**
- **Note that glycine is not a good buffer at the pH of intracellular fluid or**

blood, about 7.4

- **Another important piece of information derived from the titration curve of**

an amino acid is the relationship between its net charge and the pH of the

solution

- **At pH 5.97, glycine is present predominantly as its dipolar form, fully**

ionized but with no net electric charge

The characteristic pH at which the net electric charge is zero is called the

isoelectric point or isoelectric pH, designated pI

- **For glycine, which has no ionizable group in its side chain;**
- **the pI is simply the arithmetic mean of the two pKa values**
- **$pI = \frac{1}{2} (pK_1 + pK_2)$**
- **$= \frac{1}{2} (2.34 + 9.60) = 5.97$**
- **The farther the pH of a glycine solution is from its pI, the greater the net**

electric charge of the population of glycine molecules

- At any pH below its pI, glycine has a net positive charge and will move

toward the negative electrode (cathode)

- At any pH above its pI, glycine will have a net negative charge and thus

will move towards the positive electrode (anode)

Iodine number and it's significant

The degree of unsaturation is reflected by Iodine number.

Iodine number is defined as the number of grams of iodine absorbed by 100 gm of fat.

The more the iodine number, the greater the degree of unsaturation.

Fats rich in saturated fatty acids have low iodine numbers, while fats rich in unsaturated fatty acids have high iodine numbers

The determination of iodine number is useful to the chemist in determining the quality of an oil or its

freedom from adulteration

Iodine number of cotton seed oil varies from 103 to 111.

That of olive oil from 79 to 88,

And that of linseed oil from 175 to 202

A commercial lot of olive oil which has iodine number higher than 88 might have been adulterated with

cotton seed oil

The higher is the iodine number, the more reactive, less stable, more susceptible to oxidation and rancidification is the oil or fat.

Equilibrium constant for rate of reaction

pO₂ at Altitude

At high altitude of hills and mountains, where the pO₂ is considerably lower (due to low atmospheric pressure).

- o The delivery of oxygen to the tissues is now reduced.**
- o However, after just a few hours at the higher altitude, the BPG concentration in the blood begins to rise, leading to a decrease in the affinity of hemoglobin for oxygen.**
- o This adjustment in the BPG level has only a small effect on the binding of oxygen in the lungs .**
- o But a considerable effect on the release of oxygen in the tissues shown by increased oxygen unloading at the peripheral tissues.**
- o Increase in BPG concentration in the RBC shifts the dissociation curve to the right, i.e. increased O₂ unloading.**
- o BPG concentration also increases in those conditions in which there is decreased O₂ delivery to tissues, such as in anemia and respiratory diseases.**

o As a result, the delivery of oxygen to the tissues is restored to nearly 40% of the oxygen that can be transported by the blood.

Differentiate between Thymine and uracil

Thymine is 2,4-dioxy-5- methyl-pyrimidine

- Uracil is 2,4-dioxypyrimidine

T and U differ by only one methyl group, which is present on T but absent on U

MCQs.....

My MCQs Collecting partners 🐝....."The queen bees



1. The Trans form of maleic acid is called_____. **Fumaric Acid**
 2. Benedict test formation cuprous oxide indicated by formation_____. **Red ppt**
 3. Building block of protein is_____. **Amino Acid**
 4. _____ is deletion of a larger portion of the polypeptide chain. **Sickle Cell anemia**
- Iodine number olive oil from..... **79 to 88,**

Triacylglycerol are soluble in.....ether, benzene, acetone, chloroform)

Which amino acids has two groups, the COOH..... Glycine

SEMINAL FLUID IS RICH IN.....fructose

Constituent of chondroitin....– D -mannosamine

The amount of guanine equals the amount ofcytosine

Purine bases: Adenine (A) and Guanine (G)

Branched polysaccharide are in..... Glycogen

Standard Amino acids.....20

Number of carbon in steric acid.....18

Beta-bends are short regions usually involving successive amino acid

residues.....four

There are two subclasses of..... sphingolipids:

The deleterious effects of trans fats occur at intakes of,2 to 7 g/day

mRNA comprise about of total cellular RNA. 2–5%

pKa Acetic a, pepsin, a digestive enzyme in the stomach, is maximally active at pH 2.

.....is the catabolic end product of purines in human beings.....
Uric acid

Keratin is present in _____. All Of These

Net charge of alanine in basic sol

Sugar in DNA**T,A,C,G**

Thymine , cytosine and uracil are **Pyrimidine**

6-ring of monosaccharides?.....**pyran/pyranoses**

Palmitic acid carbon....**16:0**

Rhinoceros horn, are made up of.....**proteins.. keratin**

polymers of same monosaccharide units..... **Cellulose**

These enzymes catalyze condensation reactions joining two molecules by forming C-O, C-S, C-N and C-C bonds.....**ligases**

The mammalian ribosomes contains two major nucleoproteins subunits.....(**Large 60S smaller 40S**)

Super secondary structure of proteins....**motif**

Sialic acid gives ganglioside negative charge at pH.....**7**

Trans of maleic acid **fumaric acid**

glycosidic bond formation at which carbon no..... **alpha(1,4)**

Glucose residues in amylose are linked by _____. **A (1,4) linkage**

.Omega 6 other name **pro-inflammatory**

Ribonucleic acid (RNA)—the ..of the (DNA) — that the Master plan is expressed.....**“working copies”**

About ____ of amino acid residues in the myoglobin are found in alpha helices. ____**78%**

Percentage of rRNA in cell **80%**

Serine+ palmitate= **Sphingosine**

Cholesterol in synthesize in----...

Which one is not correct..... **Guanine is equal to Thymine**

Simple arrangement of polypeptide with recurring...is ka
secondary h

Glycosidic linkage dosra glucose say konsi carbon involve hoti
ans ***C4***

Which hemoglobin is found in adults..... **hemoglobin A**

Molecular weight of water.....**18**

The T_m of DNA is influenced by _____ **All Of Above**

2. Purine base found in RNA is_____ Guanine

3. Compounds that generate nitrous acid (such as nitrites, nitrates, and nitrosamines) change DNA

molecules by: _____ **Breakage Of Phosphodiester Bonds**

4. Under appropriate conditions DNA will form a hybrid with a complementary ____ **Both**

5. DNA model was presented by Watson and Crick in
_____ **1953**

6. Around structural core of the nucleosome. a segment of the DNA double helix is wound nearly twice -

approximately _____ **140bp**

7. Adenosine 3'-phosphate-5'-phosphosulfate is the -----
donor_____ **Sulphate**

8. A short length of DNA molecule has 80 thymine and 80 guanine bases. The total number of nucleotide

in the DNA fragment is _____ **320**

9. Which of the following is a palindromic sequence? **GGATCC CCTAGG**
10. The complex folding of large chromosomes within eukaryotic chromatin and bacterial nucleoids is generally considered ____ Tertiary **Structure**
11. Higher GC base pairs increases T_m , because , with three hydrogen bonds, require-.... heat energy to dissociate than AT base pairs. ____ **More**
12. Chromatin is composed of ____ **Nucleic Acids And Protein**
13. The sugar molecule present in nucleotide is ____ **Pentose**
14. Eukaryotic DNA is associated with tightly bound basic called histones ____ **Proteins**
15. When DNA is heated, the temperature at which ----- of the helical structure is lost is defined as the melting temperature ____ **One Half**
16. Z-DNA have a ____ **Zig-Zag Appearance**
17. RNA duplexes are -.... stable than DNA duplexes ____ **More**
18. The strand is called Template strand while ____ **3-5**
19. DNA replication is ____ **Semi-Conservative**
20. Nucleotides are composed of ____ **All**
21. Glucose residues in amylose are linked by ____ **α (1,4)** linkage
22. The number of carbon atoms in lysine is ____ **6**
23. If the carbonyl group is present at the end of the monosaccharide, then it is called ____ **Aldose**

24. The number of stereoisomers for a molecule containing only one chiral carbon is ____ **2**

25. Seminal fluid is rich in _____. Fructose

26. Which of the following glycosidic linkage exists between two glucose units of maltose? ____ **Alpha 1,4**

27. Amino sugar is formed by the replacement of the hydroxyl group at ----- of the parent monosaccharide

with amino group. ____ **C-2**

28. Which one of the following is a structural homopolysaccharide? ____ **Cellulose**

29. Which one of the following amino acids is not specified by three letter codons? **Selenocysteine**

30. Acid dissociation constants are designated as _____. **PKa**

31.. When linked to vitamins nucleotides are structural components of several essential coenzymes. For

_____ **Coenzyme A**

Remember me in your prayers...

Regards.....

Maryam Saleem khawaja...   **MERRY**  